

Project 1: Differentiated Demand for Cars

Due: October 22nd at 17:00

1 Empirical Part

1.1 Data

The dataset, `cars.csv`, has one row for $M = 5$ countries over $T = 30$ years, $t = 1970, \dots, 1999$. Throughout, we will refer to a country-year pair as a “market,” indexed by $i = 1, \dots, N$, where $N = MT = 150$. From the raw underlying dataset, the $J = 40$ highest-selling cars have been selected; that is, there are always precisely $J = 40$ discrete car alternatives available in any market i . Thus, there are $NJ = 6,000$ rows in the dataset `cars.csv`, or $TJ = 1,200$ rows per country.

Appendix A presents a full list of variables with labels and the mean values (for numeric variables). Perhaps the most important variables to consider include the price (`pr`), car weight (`we`), fuel efficiency (`li`), quantity sold (`qu`), and the country’s population (`pop`).

1.2 Questions

The overarching goal of this problem set is to imagine that you are working for the Competition Authority, having just received a formal application from VW to merge with Fiat. You must decide whether to permit this merger. Thus, the research question is:

- **Overall question:** *What would be the effects on prices and consumer welfare from a merger between VW and Fiat?*

To answer this question, please proceed along the following lines:

- **Question 1:** Estimate a simple and an extended logit model of the consumer demand for cars. Report parameter estimates, but also give the reader a sense of the own and cross price elasticities implied by your model.

Here, “extended logit” is taken to mean a more advanced model than the simple one. This can be: Nested logit, random coefficients logit, using instrumental variables, or just adding fixed effects. Note that you do not have to address every concern covered in the course (endogeneity, heterogeneity, etc.). You must choose and present arguments for those choices.

- **Question 2:** Conduct the merger counterfactual for both your logit models, and analyze similarities and differences. Compare the effects of the merger for different markets. How large would cost-savings have to be to make the merger net positive?

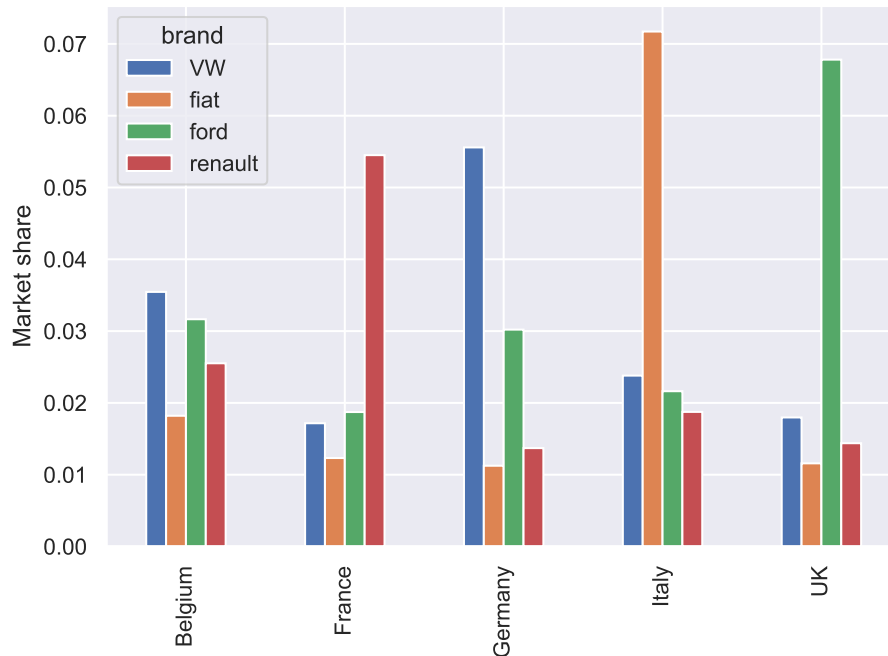
In setting up your model, you will have to make a number of practical decisions:

- **Market share:** The variable `qu` gives the quantity of sold units, and `pop` is the total population of the country (`ma`) in the given year (`ye`). However, there are several ways of defining a market share depending on what one views as the “market size.” Is the market the full population? Just the driving-age population? Individuals or households?
- **Price variable:** We recommend using the log of the price, but you could also use the price in level, or normalized by income.
- **Controls:** What variables to include in $\mathbf{x}_{jt}$: we recommend including the car’s weight (`we`), fuel efficiency (`li`), and fixed effects for the brand (`brd`). But there are many others you can choose from.
- **Instrument(s):** if you choose to use instrumental variables (IVs), there are variables relating to the exchange rates, as well as the VAT rate (`tax`). You can also construct the “differentiation IVs” (ala BLP or Gandhi & Houde).

You are also encouraged but not required to include a robustness check for one of these. Remember to argue (briefly) for your choices!

As inspiration for your analysis, you may for example choose to emphasize some notion of a “home market bias.” Figure 1 shows a simple descriptive graph that motivates why we might suspect home bias to be strong: the figure shows the market shares of four selected car manufacturers in each of the five countries from the dataset. A clear pattern emerges: each manufacturer has by far the strongest sales in its home market. Belgium serves as a neutral comparison since that country has no domestic car manufacturing.

Figure 1: Home Bias and Car Market Shares



2 Theoretical Questions

- (1) Consider the theoretical models of price competition with homogeneous products seen so far in the course. Can any of these models (perhaps with slight adjustments) potentially rationalize the observed market shares from the data for this project? If so, which model(s)? Explain and justify your answer verbally, i.e. you are not expected here to run any regressions.
- (2) Consider the Perloff-Salop model of price competition with horizontally-differentiated products, as seen in Lecture 1 and Problem Set 1. Summarize what is/are the assumed source(s) of product differentiation in this theoretical model. Then describe the assumed sources of product differentiation in the empirical model that you implemented for this project, commenting on any similarities or differences.
- (3) Test empirically whether you can reject the Perloff-Salop model (perhaps with slight adjustments regarding e.g. error terms, compared to the presentation in Lecture 1) using the data for this project.

Hint: It is sufficient to conduct the test using a joint test in a regular linear regression.

- (4) Now consider a counterfactual situation where the number of firms selling cars is very large. Based on insights from the Perloff-Salop model, what do you expect would happen to prices in this situation? Would your answer differ depending on whether ‘taste shocks’ were drawn from a normal distribution (Gaussian), instead of a Type I Extreme Value distribution (Gumbel)? If so, comment on how this difference might affect the way you interpret the coefficients obtained from the kind of empirical analysis carried out for this project.

3 Formal Requirements

- **Format:** The report must be written in English and uploaded to Peergrade via Absalon as a single PDF file.
- **Size:** Maximum 6 pages + 2 pages of output.
 - A page has fontsize = 12pt, line spacing = 1.5, and page margins 2.5 cm. Output includes tables and figures and must be formatted and labelled appropriately. I.e. make sure to choose an appropriate number of decimals and to use sensible variable labels.
 - *Note:* There is no requirement to hit the maximum number of pages. For example, 5 pages may be sufficient. Use your judgement.
- **Feedback:** After the submission deadline, you are expected to read at least one other student/group’s project and provide them with written feedback. Please comment both on what is done well and what can be clarified or improved. Be constructive!
 - **Deadline:** The deadline for providing this feedback is **Tuesday October 29 at 17.00**. Providing this feedback is a formal requirement to qualify for the final exam.
- **Code:** You must submit your Python code alongside your report. Make sure that your code runs without modifications as far as possible.¹
- **Groups:** You are allowed (and strongly encouraged) to work in groups of up to three people. List all group members on the front page of your report in alphabetical order of surnames.

¹For instance, make sure to use relative paths (i.e. not `C:\user\long\path`, but `./input/cars.csv`.)

- **AI tools:** Use of AI tools is allowed when working on and writing up the project. If you use AI tools, then you must include a brief explanation in the project as to how these tools were used.² We do not recommend directly including text in the project that was generated (solely or mainly) by an AI tool. However, if you choose to include such text, then the tool must be quoted as a source.

²For example: “I have used Github copilot to get suggestions for code, and I have made ChatGPT write first drafts for paragraphs for text based on my prompt, which I have used as a starting point for writing paragraphs of my own.”

A Variable Labels

Variable	Label	Mean
ye	year (=first dimension of panel)	84.50
ma	market (=second dimension of panel)	3.00
co	model code (=third dimension of panel)	207.50
zcode	alternative model code (predecessors and succe...	177.76
brd	brand code	16.79
type	name of brand and model	NaN
brand	name of brand	NaN
model	name of model	NaN
org	origin code (demand side, country with which c...	2.72
loc	location code (production side, country where ...	5.17
cla	class or segment code	2.30
home	domestic car dummy (appropriate interaction of...	0.32
frm	firm code	14.50
qu	sales (number of new car registrations)	35606.68
cy	cylinder volume or displacement (in cc)	1337.09
hp	horsepower (in kW)	50.10
we	weight (in kg)	934.49
pl	places (number, not reliable variable)	4.88
do	doors (number, not reliable variable)	3.55
le	length (in cm)	409.24
wi	width (in cm)	163.44
he	height (in cm)	140.46
li1	measure 1 for fuel efficiency (liter per km, a...	6.59
li2	measure 2 for fuel efficiency (liter per km, a...	8.11
li3	measure 3 for fuel efficiency (liter per km, a...	8.92
li	average of li1, li2, li3 (used in papers)	7.87
sp	maximum speed (km/hour)	154.22
ac	time to acceleration (in seconds from 0 to 100...	16.27
pr	price (in destination currency including V.A.T.)	2608988.58
prnc	=pr/(ngdp/pop): price relative to per capita i...	0.76
eurpr	=pr/avdexr: price in common currency (in SDR t...	7256.92
exppr	=pr/avexr: price in exporter currency	600384.65
avexr	av. exchange rate of exporter country (exporte...	229.30
avdexr	av. exchange rate of destination country (dest...	319.91
avcpr	av. consumer price index of exporter country	492.53
avppr	av. producer price index of exporter country	671.77
avdepr	av. consumer price index of destination country	77.02
avdppr	av. producer price index of destination country	87.69
xexr	avdexr/avexr	66.32
tax	percentage VAT	0.21
pop	population	49183800.00
ngdp	nominal gross domestic product of destination ...	178667304825541.97
rgdp	real gross domestic product	216716720230172.44
engdp	=ngdp/avdexr: nominal gdp in common currency (...)	504371708122.45
ergdp	=rgdp/avexr	659002565347.56
engdpc	=engdp/pop: nominal gdp per capita in common c...	10015.84
ergdpc	=ergdp/pop	13264.84
s	market share (qu / qu_tot)	0.02
qu_tot	total sales in this market-year	1424267.29
inc	avg. income per capita	26829.74